claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ab7c5e07ea26ec571.jsonl`
- SHA-256: `9a047bff98cf6c0f8760258e86e3e04a44019f8c098800e2acbb729a82653480`
- Source modified: `2026-06-16T03:21:00+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:20:03.177Z)

Source Extractor

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Research question: "Research question: **What concrete, recent (2022?2025) techniques and
evaluation practices should a local, commercially-licensed badminton rally-detector adopt to
close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16
golden videos)?** Produce an adversarially-verified, cited report organized around the four
dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over
generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a
heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff"
confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits
far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it
does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active"
iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold;
active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary
split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves +
knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-
frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom
licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM
(Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference;
minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture
skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) +
late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet,
See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation,
tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-
one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-
count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE,
a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment
leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal
windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-
window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ?
what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented
actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression
and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression),
BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads),
smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,
```

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:
URL: https://arxiv.org/pdf/2012.08668
Title: Mitigating Bias in Calibration Error Estimation (Roelofs et al., AISTATS 2022)
Found via: Small-n evaluation rigor for video segmentation search

Task
1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
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   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:20:05.709Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:20:19.327Z)

I've analyzed the provided PDF content, but I must note that the document appears to be heavily compressed/encoded binary data (a PDF stream) rather than readable text. The content shown consists primarily of:

- PDF object definitions (streams, form XObjects, filters)
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"calibration_fig3.pdf")

Unfortunately, I cannot extract the specific claims you requested because the actual text content of the paper is not decodable from this raw PDF structure. The readable elements are limited to file references and structural metadata.

To properly answer your questions about:
1. Bias in standard ECE estimators
2. Recommended alternative estimators
3. Guidance on bin selection and debiasing
4. Quantitative results and theorems
5. Publication venue and date

I would need either:
- The extracted/converted text version of this paper
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- The paper's DOI or direct link to access it through standard academic channels

Recommendation: Could you provide the paper in text format, or share its title/authors so I can reference the published version directly?

[Binary content (application/pdf, 1.7MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781580013109-x8y2q3.pdf]

4. assistant (2026-06-16T03:20:22.071Z)

The PDF binary wasn't decodable via WebFetch. Let me try the arXiv abstract/HTML page instead, and also read the saved PDF directly.

5. user (2026-06-16T03:20:25.275Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

6. user (2026-06-16T03:20:29.648Z)

Analysis of Calibration Error Estimation Paper

Abstract & Key Claims

Main Finding on Bias:
The paper identifies that "binning-based estimators with bins of equal mass (number of instances) have lower bias than estimators with bins of equal width."

Recommended Estimators:
Two reliable approaches are proposed:
1. The debiased estimator (existing method from Brocker 2012; Ferro & Fricker 2012)
2. ECE_sweep (novel method), which "uses equal-mass bins and chooses the number of bins to be as large as possible while preserving monotonicity"

Specific Claims About Bins & Debiasing:
The abstract states these estimators yield "improvements in the effectiveness of recalibration methods and in the detection of model miscalibration," though specific numerical reductions are not quantified in the abstract itself.

Methodology:
The framework "involves synthesizing model outputs that have the same statistics as common neural architectures on popular data sets" to assess bias computationally.

Publication Details:
- **Venue:** AISTATS 2022
- **Submission Date:** December 15, 2020
- **Final Version:** February 11, 2022 (v3)

- **arXiv ID:** 2012.08668

Code Availability: Implementation available at the referenced GitHub repository (caltrain).

7. assistant (2026-06-16T03:20:33.538Z)

The arXiv abstract page fetch succeeded and gave me the key claims. Let me try to extract more detail from the PDF text body to verify specific quotes and numbers, using a Python-based text extractor.

8. user (2026-06-16T03:20:37.208Z)

Mitigating Bias in Calibration Error Estimation
Rebecca Roelofs Nicholas Cain Jonathon Shlens Michael C. Mozer
Google Research Google Research Google Research Google Research
Abstract
For an AI system to be reliable, the con?-
dence it expresses in its decisions must match
its accuracy. To assess the degree of match,
examples are typically binned by con?dence
and the per-bin mean con?dence and accuracy
are compared. Most research in calibration
focuses on techniques to reduce this empiri-
cal measure of calibration error,ECEbin. We
instead focus on assessing statistical bias in
this empirical measure, and we identify better
estimators. We propose a framework through
which we can compute the bias of a partic-
ular estimator for an evaluation data set of
a given size. The framework involves syn-
thesizing model outputs that have the same
statistics as common neural architectures on
popular data sets. We ?nd that binning-based
estimators with bins of equal mass (number
of instances) have lower bias than estimators
with bins of equal width. Our results indi-
cate two reliable calibration-error estimators:
the debiased estimator (Brocker, 2012; Ferro
and Fricker, 2012) and a method we propose,
ECEsweep, which uses equal-mass bins and
chooses the number of bins to be as large as
possible while preserving monotonicity in the
calibration function. With these estimators,
we observe improvements in the e?ectiveness
of recalibration methods and in the detection
of model miscalibration.
1 INTRODUCTION
Machine learning models are increasingly deployed in
high-stakes settings like self-driving cars (Caesar et al.,
2020; Geiger et al., 2013; Sun et al., 2020) and medical
diagnosis (Esteva et al., 2017, 2019; Gulshan et al.,
2016) where it is critical to recognize when a model is
likely to be incorrect. Unfortunately, models often fail
Proceedings of the 25thInternational Conference on Arti?-
cial Intelligence and Statistics (AISTATS) 2022, Valencia,
Spain. PMLR: Volume 151. Copyright 2022 by the au-
thor(s).
in unexpected and poorly understood ways, hindering
our ability to interpret and trust such systems (Azulay
and Weiss, 2018; Biggio and Roli, 2018; Hendrycks
and Dietterich, 2019; Recht et al., 2019; Szegedy et al.,
2013). To address these issues, calibration is used to
ensure that a model produces con?dence scores that
re?ect its ground truth likelihood of being correct (Platt
et al., 1999; Zadrozny and Elkan, 2001, 2002).

To obtain an estimate of the calibration error, or ECE1, the standard procedure partitions the model con?dence scores into bins and compares the model?s predicted accuracy to its empirical accuracy within each bin (Guo et al., 2017; Naeini et al., 2015). We refer to this speci?c metric asECEbin. Recent work observed that the calculation ofECEbin is sensitive to implementation (Kumar et al., 2019; Nixon et al., 2019). Fundamentally, akeyconfoundingfactoris statistical bias, thedi?erence between the expectedECEbin and the true calibration error (TCE). Because bias is largely unexplored in the literature, its magnitude and sign is unknown, as is its dependence on hyperparameters of theECEbin estimator (e.g., number of bins, how bins are formed). We explain our reasons for focusing on estimator bias and not variance in Section 4.

Bias inECEbin measurement has two real world conse- quences. First, the measurement of calibration error on a given model may be systematically incorrect. Thus, our understanding of how well a model knows whether it is correct may be poor, and may not be accurately captured by naively reportingECEbin. Second, many techniques have been developed to minimize the cali- bration error, such as post-hoc recalibration techniques (Guo et al., 2017; Zadrozny and Elkan, 2001, 2002) and, more recently, calibration-sensitive training objectives (Karandikar et al., 2021; Krishnan and Tickoo, 2020; Kumar et al., 2018; Lin et al., 2018; Mukhoti et al., 2020). Given that the selection of the training objec- tives and the justi?cation of a recalibration technique is predicated on the measurement of the calibration

1Naeini et al. (2015) introduce ECE as an acronym for Expected Calibration Error. However, ECE is not a proper expectation whereas thetrue calibration erroris computed under an expectation. To resolve this confusion, we prefer to read ECE asEstimated Calibration Error.

arXiv:2012.08668v3 [cs.LG] 11 Feb 2022Mitigating Bias in Calibration Error Estimation

0.0 0.2 0.4 0.6 0.8 1.0
Predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0Empirical accuracy
Sample B
Raw data
True calibration
ECE Bins
0 5 10 15 20
ECEBIN (%)
0.00
0.05
0.10
0.15
0.20PDF
ECEBIN Distribution
Mean
Sample B

(a) Figure 1: ECEbin exhibits large bias for perfectly cali- brated models. We simulate data from a perfectly calibrated model with con?dence scores ?t to ResNet-110 CIFAR-10 output(Heetal.,2016;Kängsepp,2019)andmeasure ECEbin using 15 equal-width-spaced bins. The left panel shows a re- liability diagram for a sample of sizen = 200 (named Sample

B); the right panel has a distribution ofECEbin scores com-
puted across 106 independent simulations. Even though the
model is perfectly calibrated, ECEbin systematically predicts
large calibration errors.
error, reliance on an inaccurate estimator may lead to
a suboptimal choice.
We address this problem by developing a technique
to measure bias in calibration metrics, which we call
the bias-by-construction (BBC)framework. The BBC
framework uses simulation to create a setting where the
TCE can be computed analytically and thus the bias
canbeestimateddirectly. BBCrevealsthat ECEbin has
systematic non-negligible statistical bias, particularly
for perfectly calibrated models (Figure 1).
Our goal is to identify the least biased estimator of cal-
ibration error using BBC. We consider two estimators
previously proposed in the literature: thedebiased esti-
mator (Brocker, 2012; Ferro and Fricker, 2012), which
we refer to as ECEdebias, and the smoothed kernel
density

9. assistant (2026-06-16T03:20:39.992Z)

I now have the verbatim abstract and introduction text with strong, falsifiable claims. I have enough to extract the claims with direct quotes.

10. user (2026-06-16T03:20:50.499Z)

Structured output provided successfully

11. assistant (2026-06-16T03:20:59.928Z)

I've extracted and verified the claims from the source. The fetch succeeded (verbatim abstract plus introduction text confirmed directly from the PDF body), so this is a primary research source.

Summary: Roelofs et al. (AISTATS 2022), "Mitigating Bias in Calibration Error Estimation," is directly relevant to Dimension 4 (evaluation rigor at small n). Key takeaways for the badminton rally-detector leaderboard:

- Plain binned ECE has systematic, non-negligible bias that over-reports calibration error, worst for well-calibrated models ? so reporting raw ECE on ~6?16 videos is untrustworthy.
- Use equal-mass bins (not the default equal-width) for lower bias.
- Adopt one of two reliable estimators: the debiased estimator (Brocker 2012; Ferro & Fricker 2012) or their ECE_sweep (equal-mass bins, max bins preserving monotonicity).
- The bias is acute at small evaluation sizes (their demo uses n=200), reinforcing that the current Platt/isotonic + Brier/ECE harness should switch off naive equal-width ECE.

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